

HR Interview Scoring Engine

Overview

The HR Interview Scoring Engine evaluates candidate responses during an HR interview using multiple scoring dimensions. Instead of relying on a single metric, the engine combines answer relevance, communication quality, confidence indicators, and response consistency to generate a structured interview score.

This component builds on the existing ZecPath AI scoring modules and provides a unified evaluation layer for HR interview responses.

Objectives

The engine is designed to:

- Evaluate how relevant an answer is to expected interview topics.
 - Integrate communication quality analysis.
 - Integrate confidence analysis.
 - Measure consistency between candidate responses.
 - Calculate a weighted score for each answer.
 - Generate an overall interview score.
 - Classify interview performance into understandable levels.
 - Generate a structured candidate HR report.
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Scoring Components

The engine evaluates four major dimensions.

Answer Relevance

Answer relevance measures how closely a candidate response matches expected keywords or topics. The relevance score is calculated using keyword overlap between the candidate answer and expected keywords.

A response containing all expected keywords receives a higher relevance score, while a response containing only some expected keywords receives a proportional score.

Communication Score

Communication scoring is integrated from the existing CommunicationScoringEngine.

The communication evaluation considers:

- Fluency
- Grammar
- Vocabulary
- Clarity
- Filler words

- Response structure

The resulting communication score becomes one component of the HR interview evaluation.

Confidence Score

Confidence evaluation is integrated from the existing ConfidenceSentimentEngine.

The confidence analysis considers signals such as:

- Hesitation
- Response length
- Pace when duration is available
- Sentiment
- Uncertainty
- Consistency signals

These indicators are combined to generate a confidence score.

Consistency Score

Consistency scoring evaluates how consistent a candidate answer is with a previous answer when previous interview context is available.

For the first answer, where no previous response exists, the consistency score defaults to 100 because there is no earlier response available for comparison.

Weighted Score Calculation

The final answer score is calculated using configurable weights.

The Day 37 configuration uses:

Component	Weight
Answer Relevance	35%
Communication	25%
Confidence	25%
Consistency	15%

The weighted score combines all four scoring dimensions and normalizes the final result to a range between 0 and 100.

Score Classification

Interview scores are classified using configured thresholds.

Score Range	Classification
80 and above	Excellent
65 to 79	Good
50 to 64	Moderate
Below 50	Needs Improvement

These classifications make numerical scores easier to interpret during candidate evaluation.

Single Answer Evaluation

The `score_answer()` method evaluates an individual candidate response.

The result includes:

- Original answer
- Individual component scores
- Weighted score
- Communication analysis details
- Confidence analysis details

This allows the system to retain both the final numerical score and the underlying evaluation signals.

Interview-Level Evaluation

The `score_interview()` method evaluates multiple interview responses.

For each answer, the engine:

1. Calculates answer relevance.
2. Evaluates communication quality.
3. Evaluates confidence.
4. Evaluates consistency.
5. Generates a weighted answer score.

The individual results are aggregated to calculate an overall interview score.

The interview result includes:

- Answer count
- Individual answer results
- Interview score
- Interview level
- Score breakdown

The interview score uses average-per-answer normalization.

Candidate Report Generation

The `generate_candidate_report()` method produces a structured HR evaluation report.

The report includes:

- Overall interview score
- Interview performance level
- Number of answers evaluated
- Score breakdown
- Answer-level results
- Normalization method

The normalization method is:

`average_per_answer`

This makes the scoring approach transparent and consistent across interviews with different numbers of questions.

Input Validation

The engine validates important input conditions.

Examples include:

- Interview answers must be provided as a list.
- Required answer fields must be present.
- Invalid input types raise appropriate errors.
- Empty interviews return a valid zero-score result.

These checks improve reliability when the engine is integrated into larger interview workflows.

Integration Architecture

The Day 37 engine integrates existing ZecPath AI modules.

Candidate Answer | v HRInterviewScoringEngine | +---- Answer Relevance | +----
CommunicationScoringEngine | +---- ConfidenceSentimentEngine | +---- Consistency Analysis | v Weighted
Answer Score | v Interview Score | v Candidate HR Report

This architecture avoids duplicating existing scoring logic and provides a unified HR interview evaluation layer.

Testing

A dedicated Day 37 test suite was created for the HR Interview Scoring Engine.

The tests cover:

- Configuration loading

- Engine initialization
- Full keyword relevance
- Partial keyword relevance
- Empty answer handling
- Answers without expected keywords
- First-answer consistency
- Multi-answer consistency
- Single answer scoring
- Interview scoring
- Empty interview handling
- Invalid input validation
- Missing answer validation
- Score classification
- Candidate report generation

Day 37 Test Result

15 passed

Full Regression Testing

After completing Day 37 testing, the complete ZecPath AI project regression suite was executed.

Result:

183 passed 2 warnings 0 failures

The warnings were existing project warnings and were unrelated to the Day 37 implementation.

This confirms that the HR Interview Scoring Engine integrates successfully without breaking existing project functionality.

Conclusion

The Day 37 HR Interview Scoring Engine provides a structured and extensible evaluation layer for HR interview responses.

By combining relevance, communication, confidence, and consistency into a weighted scoring system, the engine produces both answer-level and interview-level evaluations.

The module also generates structured candidate reports and integrates with existing ZecPath AI scoring components, supporting the development of a more complete AI-powered interview evaluation workflow.